

as a load on the TFT. (A liquid-crystal capacitor is a liquid crystal that acts as a capacitive load.) A pixel is selected by applying pulse signal V_{SEL} to particular gate line G_n from the gate driver. At the same time, data signal in the analogue positive voltage V_{DAT} is transferred to the drain of the transistor from the source driver. When the TFT is 'on', the source and the drain short, which directly applies V_{DAT} to the liquid crystal and the storage capacitor (C_s). C_s charges and sustains the charge when the TFT is 'off'. The main function of C_s is to maintain voltage across the liquid crystal until the next select line voltage is applied for refreshing the picture.

Active addressing of a 3×3 matrix in active-matrix LCD

The TFT-LCD panel of the AMLCD is scanned sequentially line-by-line from top to bottom. Each line is selected by applying a pulse of +20V to gate line G_n , which turns on the TFTs in that specific row. Rows are deselected by applying -5V to G_{n+1} and G_{n-1} , which turns off all the TFTs in the deselected rows and then the data signal is applied from the source driver to the pixel electrode.

The voltage applied from the source driver, called 'gray-scale voltage,' decides the luminance of the pixel. The storage capacitor (C_s) maintains the luminance of the pixel until the next frame signal voltage is applied. In this way, the next line is selected to turn on all the TFTs, then the data signal is fed from the source driver and hence scanning is done.

TFT-LCD panel controller/driver

The TFT-LCD controller/driver unit consists of a TFT-LCD timing controller with low-voltage differential signaling (LVDS) receiver, reduced-swing differential signaling (RSDS) transmitter, oscillator, gray-scale reference voltage generator, gate driver, and source or column driver as shown in Fig. 9.

For large panel size, high resolution and high gray-scale, high-speed

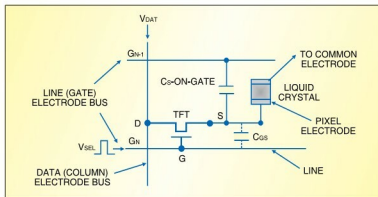


Fig. 6: Equivalent circuit of a unit pixel

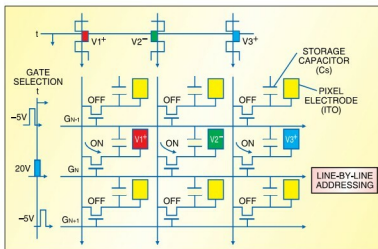


Fig. 7: Active addressing of 3×3 matrix

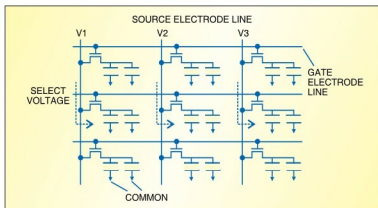


Fig. 8: Equivalent circuit of 3×3 matrix shown in Fig. 7

data transmission is required between the TFT-LCD timing controller and the source driver. To achieve high-speed data transmission, low-voltage techniques such as mini-LVDS and RSDS having differential voltage of 350 mV across the 100-ohm termination resistor at clock frequency of 85 MHz to 170 MHz are used. The LVDS receiver transfers data serially bit-by-bit over a single pair of wires. This allows fast

pixel individually in sync with the gate driver and source driver and, at the same time, the RGB video signal is passed through the source driver to TFT-LCD cells. The gray-scale reference generator generates reference voltages to convert the digital RGB signal into analogue RGB signal using either a voltage divider or DC-DC converter. The gate driver selects the TFTs by applying a positive pulse to

data transfer with less power consumption.

The LVDS receiver receives 24-bit video RGB data and control signal through eight pairs of LVDS channels and two pairs of LVDS clocks.

The timing controller generates various control signals to drive the gate driver and the source driver in order to produce images on the screen.

As mentioned earlier, in a TFT-LCD panel, scanning is done line-by-line from top to bottom and each pixel is addressed by selecting the gate line and applying the data signal to TFT-LCD cells through the source driver. The controller generates control signals to address each